

Offense vs. Defense:

A Regression Analysis of What Predicts NBA Team Success

I. Introduction

Throughout the history of the NBA, the league has gone through multiple periods of evolution. Commonly, these evolutionary periods are sparked by player signings, draft picks, and rules changes. A particularly interesting development in the league is the popularity of three pointers. Since the 2000s, three point attempts across the league have more than doubled. Along with this, the league also saw an increase in offensive efficiency and creativity. As a result, statistics and analytics are now crucial to these teams. While coaches and front offices can utilize these statistics to shape rosters and strategies, analysts and fans can use such analytics to create engaging debates.

As the games and players have evolved, so has the statistics that are being measured. This leads to a wide range of efforts and research to identify key metrics that are linked to winning. Oliver (2004) dedicated the “Four Factors” which are effective field goal percentage, offensive rebounding rate, free throw rate, and turnover rate to be most linked to a team’s effectiveness. More recently, shot selection, spacing, and three point efficiency have also been credited for the team’s success by Goldsberry (2019). In spite of such results, there has not been an agreed upon statistics or combination of statistics that accurately predict a team’s success aside from the obvious variables such as shot efficiencies. Specifically, the goal of this research is to examine a comprehensive set of team statistics over a long historical window and dive into the relationship between a team’s offensive and defensive performance and the teams’ outcomes across multiple seasons. To achieve this goal, this project will study the following question: Do offensive or defensive metrics contribute more to NBA teams performance from 2000 to 2023?

II. Methodology

A. Data

The dataset discussed above was originally uploaded to Kaggle in 2023. Michael H., the author of this dataset, scraped team statistics directly from [NBA.com](https://www.nba.com) to examine the NBA teams' performances across each regular season and playoff run from the 2000-01 through 2023-24 seasons. This dataset, as a whole, includes 716 observations and 29 variables with no missing values. Each row represents an individual NBA team and its different statistics and performance in a particular season. The columns include team identifiers (team, season), total results (games and minutes played, wins, losses, win percentage), along with other meaningful statistics.

Some offensive rating variables are makes, attempts, and percentage for field goal, three point, and free throws. To illustrate shot selection and efficiency, these metrics are combined with total points scored. Other common variables include offensive/defensive/total rebounds, assists, turnovers, steals, blocks, and fouls. A more complex variable includes plus minus which encapsulates a team's scoring performance in comparison to their opponents' during certain stretches of the game.

To make sure that our model captures impactful variables for team successes, the analysis will mainly focus on certain performance efficiency metrics. Variables that transparently and directly affect wins like wins, losses, win rate, and plus minus will be excluded. In contrast, variables such as metrics concerning free throws, three points, and field goals will be included. To explore further, statistics like assists, turnovers, rebounds, steals, blocks, and fouling will also be considered. In selecting metrics that are efficiency, offensively, and defensively based, the model will be able to identify driving factors to a team's success while keeping in mind the

various aspects of the game. To further characterize, the offensive metrics that will be considered are shooting percentages, assists, turnovers, and offensive rebounds. For defensive metrics, steals, blocks, fouls committed, and defensive rebounds will be considered.

B. Model

The response variable in this study is wins, which represents the number of wins a NBA team won in a particular season. It is a quantitative and numeric measure that isn't mathematically considered continuous as it is an integer on a 0 to 82 scale. For the purpose of this study, due to the reasonably large interval, wins behave like a continuous variable. As a result, multiple linear regression models have been chosen for this study. That leaves us with the following two models for offensive and defensive metrics:

- $wins = \beta_{0,O} + \beta_{1,O}field_goal_percentage + \beta_{2,O}three_point_percentage + \beta_{3,O}free_throw_percentage + \beta_{4,O}assists + \beta_{5,O}offensive_rebounds + \beta_{6,O}turnovers + \epsilon_O$
- $wins = \beta_{0,D} + \beta_{1,D}steals + \beta_{2,D}blocks + \beta_{3,D}personal_fouls + \beta_{4,D}defensive_rebounds + \epsilon_D$

The first model represents the multiple linear regression model for offensive metrics and the second model is for defensive metrics. The error term ϵ represents the noise that could be affecting wins and is under the assumption that it is normally distributed with constant variance. The data set is then randomly split into training and test sets. The training set comprised 70% of the original data and will be used to fit and refine the linear regression models.

After fitting the regression models, the next step was to evaluate possible interaction effects. As there are numerous variables to consider, contextual plausibility was used. Starting with the offensive metrics model, the effect of `field_goal_percentage` x `assists` were considered. As a team that can efficiently score field goals, an assist can lead to better shot selections. For the

defensive metrics model, the interaction effect of blocks x defensive rebounds was considered as blocking a shot is only really effective when the team comes up with a defensive rebound. Each interaction term was added to its respective model and an ANOVA test was used to evaluate whether the effect was significant.

Finally, the linearity, constant variance, multicollinearity, and normality assumptions were checked. A residuals vs fitted plot was made to examine the relationship between predictors and wins and whether there existed a pattern with the spread of residuals. As for the normality of residuals, a normal Q-Q plot was made and examined. Lastly, a variance inflation factor test was run to check for multicollinearity.

III. Results

1. Baseline Models

Before the two final offensive and defensive regression models were fit to estimate their influence on wins, a descriptive summary of the quantitative variables used throughout the study was generated. As can be seen, Table 1

Table 1: Descriptive Statistics for Quantitative Variables

Variables	Mean	Standard Deviation
wins (response)	40.21508	12.21502
field-goal percentage	45.57374	1.686295
three-point percentage	35.58841	1.894185
free-throw percentage	76.19888	3.002977
assists	1812.225	209.5204
turnovers	1155.024	111.1405
offensive rebounds	881.3492	121.5442
defensive rebounds	2557.714	205.076
steals	609.4106	73.74201
blocks	393.4288	67.00249
personal_fouls	1668.816	175.6804

reports the mean and standard deviation for each variable. The result illustrates the variation in statistics between different teams across the NBA. Furthermore, all the VIF values of the predictors in both the baseline models are under 3 signalling the absence of multicollinearity.

2. Interaction Terms

Two different contextually plausible interaction effects were tested to determine whether their combined effects improved the individual model performance. Starting with the offensive

model, the added interaction term `field_goal_percentage x assists` showed improvement in comparison to the baseline model with an F score of 17.02 and p-value of 0.001. On the other hand, the interaction of `blocks x defensive rebounds` for the defensive model did not yield statistically significant improvement with an F score of 0.14 with a p-value of 0.706. As a result, only the interaction term for the offensive model will be added.

3. Final Models Summary

Until now, the offensive model included field-goal and three-point percentages, offensive rebounds, turnovers, and the interaction term of field-goal percentage and assist. Free throw percentage was removed as it was insignificant in the baseline model. The reason why assists, which was also insignificant, was not removed as we are investigating its interaction with field goal percentage. After model refinement, the

interaction term was not found to be statistically significant under any reasonable values of

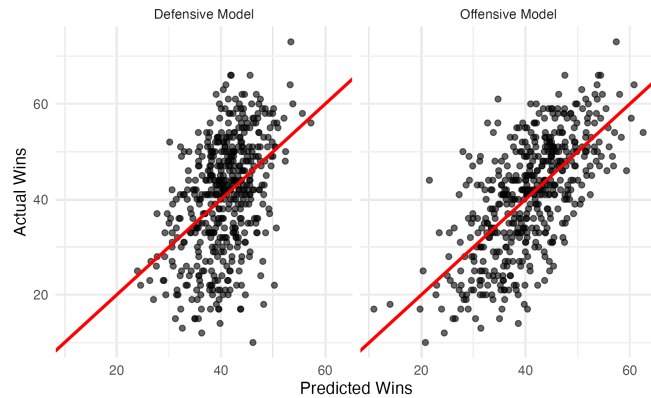
alpha. As a result, the final offensive model achieved an adjusted R^2 of 0.425 is as follows:

$wins = \beta_{0,0} + \beta_{1,0}field_goal_percentage + \beta_{2,0}three_point_percentage + \beta_{3,0}offensive_rebounds + \beta_{4,0}turnovers + \epsilon_0$. As for the final defensive model, it is exactly as the proposed

Table 2: Parameters Estimates

Predictors	Estimate	Standard Error	P-value
Intercept (O)	-183.60	14.38	<0.001
Field-Goal Percentage (O)	3.444	0.307	<0.001
Three-Point Percentage (O)	1.772	0.267	<0.001
Offensive Rebounds (O)	0.03533	0.00406	<0.001
Turnovers (O)	-0.02354	0.00411	<0.001
Intercept (D)	-19.858	7.890	0.012
Steals (D)	0.03187	0.00726	<0.001
Personal Fouls (D)	-0.00932	0.00301	0.002
Blocks (D)	0.03492	0.00766	<0.001
0.01674	0.01674	0.00249	<0.001

Figure 1: Predicted vs Actual Wins



baseline model: $wins = \beta_{0,D} + \beta_{1,D}steals + \beta_{2,D}blocks + \beta_{3,D}personal_fouls + \beta_{4,D}defensive_rebounds + \varepsilon_D$.

4. Model Assumptions

As stated before, with vif values under 3, there is no sign of multicollinearity in both of the models. From the diagnostics plots of the final offensive and defensive models can be examined to assess the model assumptions. In both models, there are no violations of linearity, normality, or homoscedasticity. The Residuals vs. Fitted plot illustrates no clear trend or curvature and the Normal Q-Q plot shows only mild tail deviation. It is consistent that the plots of the defensive model show slightly greater spread of residual and tail deviation due to its weaker explanatory power.

IV. Discussion

The results presented above argue that offensive efficiency, in comparison to defensive metrics, provides greater predictability for wins. From figure 1, it is apparent that the offensive model with field-goal and three-point percentages, offensive rebounds, and turnovers explain more of the variation in a team's performance than that of the defensive model. The offensive model's prediction aligns closely with the red reference line, illustrating its performance with capturing average wins total and identifying between high and low performing teams. The defensive model's vertical scatter is consistent with its inaccuracy in predictions and larger residual errors. This result reinforces that a team that can generate efficient and effective offensive strategies will gain more advantages to its performance than if they focused on defensive statistics. Despite also having highly statistical significance predictors, the defensive model explains a lot less of the variation in the data. This is more likely a commentary on the lack of descriptive defensive statistics as a lot of defensive efforts in a game is immeasurable

through numbers. A statistic that is particularly interesting is the statistical insignificance of free throw percentage in the offensive model. Nowadays, foul-baiting is a popular point of discussion among NBA analysis and fans. Some go as far as to say that foul-baiting is ruining the league as players would intentionally draw a defensive foul to shoot free throws. This is highly efficient as free throws are two uncontested one-point shots where the time is not officially running. Despite this, free throw percentage did not emerge as a significant indicator of a team's performance. This result can also indicate that the teams across the league are clustered around a similar value of free throw percentage.

V. Conclusion

Altogether with the adjusted R^2 values, significance of predictors, and prediction accuracy plot, the result indicates that offensive performance is a better and more reliable predictor of an NBA Team's performance than the defensive metrics considered in this study. For future analyses, more complex and richer defensive statistics that aren't captured by the box-score should be incorporated. Researchers may also find more success by exploring nonlinear models to explore the relationship between winning and the accompanying predictors. Finally, while numbers and data can illustrate general patterns, it is important to include contextual understanding to avoid diminishing the complexity of the different aspects of the sport.

VI. References

- Oliver, D. (2004). *Basketball on Paper: Rules and Tools for Performance Analysis*. Potomac Books.
- Goldsberry, K. (2019). *SprawlBall: A Visual Tour of the New Era of the NBA*. Mariner Books.